MLflow Experiment Tracking Report

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March 9, 2026

1 Introduction

This report presents the results of five experiment runs tracked using MLflow. Different hyperparameter configurations were explored to evaluate their impact on the model performance. The main evaluation metric used for comparison was **Discriminator Accuracy (D_accuracy)**.

2 Table View

Table 1 shows the five experiment runs sorted by the primary performance metric (Discriminator Accuracy).

Run Name	Accuracy	Loss_D	Loss_G	Epochs
industrious-stag-292	1.0000	0.0000	100.0000	10
worried-chimp-726	0.9881	0.1608	4.9441	10
merciful-dove-458	0.9881	0.0771	6.2270	10
agreeable-seal-312	0.9762	0.2714	5.8024	10
marvelous-calf-919	0.9524	0.2208	7.8582	10

Table 1: Experiment runs sorted by Discriminator Accuracy

Figure 1 shows the MLflow table view used to compare runs.

						Metrics		
Run Name	Created	Dataset	Duration	Source	Models	D_accuracy	loss_D	loss_G
agreeable-seal-312	1 day ago	-	2.4min	main.py	discriminator_epoch_... +1	0.976190447...	0.271422445...	5.802362918...
merciful-dove-458	1 day ago	-	2.2min	main.py	generator_epoch_10	0.988095223...	0.077072106...	6.226985931...
marvelous-calf-919	1 day ago	-	2.2min	main.py	generator_epoch_10	0.952380955...	0.220767199...	7.858232021...
industrious-stag-292	1 day ago	-	2.1min	main.py	discriminator_epoch_... +1	1	0	100
worried-chimp-726	1 day ago	-	2.1min	main.py	-	0.988095223...	0.160786837...	4.944058895...

Figure 1: MLflow Table View of experiment runs

3 Comparison View

To better understand the training behavior, multiple runs were selected and their performance curves were overlaid using the MLflow comparison tool. The accuracy and loss curves help visualize convergence behavior and training stability.

Figure 2 shows the comparison of at least three runs.

Figure 2: Overlayed training curves of multiple runs

4 Artifact Gallery

MLflow automatically logs several artifacts generated during the experiment. These artifacts include the trained model and metadata describing the model environment.

The artifact gallery includes:

- The saved trained model
- The `MLmodel` metadata file
- The automatically generated environment files

Figure 3 shows the artifacts tab in MLflow.

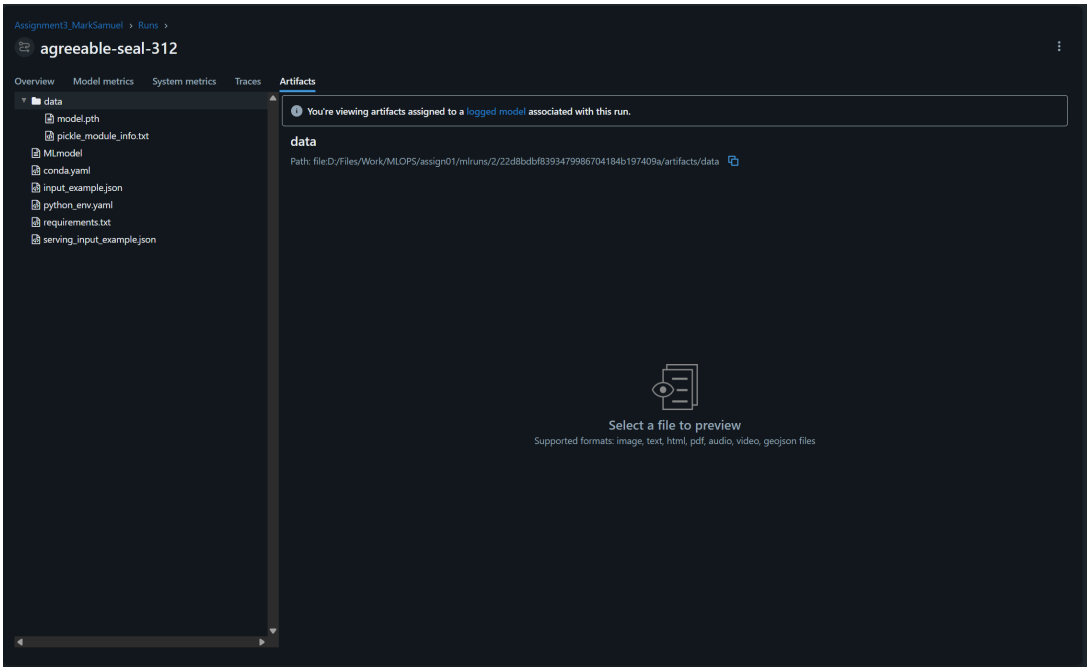

Figure 3: MLflow Artifact Gallery

5 Short Analysis

Among the five runs, the run **worried-chimp-726** was considered the best overall run. Although the run **industrious-stag-292** achieved a perfect accuracy of 1.0, its generator loss reached a very high value (`loss_G = 100`), which indicates unstable GAN training and possible mode collapse.

The run **worried-chimp-726** achieved a high discriminator accuracy (**0.988**) while maintaining a much lower generator loss compared to the perfect accuracy run. This suggests a more balanced training process between the generator and discriminator.

From the comparison curves, we can observe signs of:

- **Overfitting**: Runs where discriminator accuracy quickly approaches 1.0 while generator loss grows very large.
- **More stable training**: Runs where both losses remain moderate and accuracy improves gradually.

Overall, the selected winning run provides the best trade-off between high accuracy and stable adversarial training dynamics.

6 Conclusion

MLflow proved to be a useful tool for tracking experiments, comparing model runs, and storing artifacts. The experiment analysis demonstrated how different runs can be compared using metrics, training curves, and saved artifacts to determine the most reliable model configuration.